

ASHEN THILAKARATHNA

FULL-STACK DEVELOPER · IMMERSIVE EXPERIENCE ENGINEER

33/C, Mawathagama, Padukka (10500), Western Province, Sri Lanka. | +94 71 976 3073

ashein.me | Hello@ashein.me | [linkedin.com/in/ashein](https://www.linkedin.com/in/ashein) | github.com/asheint | medium.com/@asheint

SUMMARY

I am a full-stack developer specializing in immersive technologies, including AR, 3D web, and real-time systems. I have production experience with Unity, Three.js, NestJS, and React, building scalable, end-to-end applications. I am comfortable working across the stack, handling rapid context switching, and delivering features efficiently. I take ownership of my work, contribute effectively within teams, and also build independent products to continuously improve my technical depth.

EXPERIENCE

Associate Immersive Experience Developer

(Nov 2025 - Present)

Zenith Xperience (Pvt) Ltd · Colombo, Sri Lanka · Full-time

- **AR Garment Visualisation Platform (Augmented Reality):**
 - Built a mobile AR application using Unity 6 and AR Foundation, enabling overseas clients of a leading garment manufacturer to view, inspect, and interact with 3D garment models in augmented reality on their devices.
 - Implemented a real-time multi-user room system using Colyseus, clients in the same session see each other's presence and live annotation activity as it happens.
 - Built annotation tools allowing users to mark up and comment on specific areas of the AR garment, with updates streamed in real time to all active viewers.
 - Developed the accompanying client portal.
- **Aviation Runway Inspection Tool:**
 - Contributing to a full-stack platform for annotating defects, cracks, and maintenance jobs on airport runways, responsible for UI implementation and admin dashboard development.
- **Healthcare Platform:**
 - Full-stack contributor on a patient-practitioner connection platform covering appointment management and waitlist functionality, primarily frontend, with backend defect resolution and automated email template implementation.

Immersive Experience Developer Apprentice

(Jul 2025 – Nov 2025)

Zenith Xperience (Pvt) Ltd · Colombo, Sri Lanka · Apprenticeship

- Joined during the final delivery phase of an AR cooking instruction application built for XReal 1 smart glasses, resolved dashboard issues, and contributed to feature completions.
- Supported company social media content production and visual communications.

EDUCATION

B.Sc. in Computer Science & Physics

(May 2021 – May 2025)

University of Peradeniya · Sri Lanka

- Relevant coursework: Web Programming, Computer Architecture, Data Structures, Artificial Intelligence, Server-Side Web Programming.

GCE Advanced Level — Physical Science Stream

(Jan 2017 – Aug 2019)

D.S. Senanayake College · Colombo 07

- Z-Score: 1.3224 · Physics (B), Chemistry (B), Combined Mathematics (C), General English (B).

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PROJECTS

fansoundzzz.com — Immersive Ambient Sound Platform [↗](#)

Personal · Live

- Built the platform with a fully immersive 3D web environment using Three.js and WebGL, providing atmospheric ambient sound spaces for focus, work, and sleep.
- Handles audio layering and scene interaction as well.

prompt.io — AI Prompt Marketplace [↗](#)

Personal · Live (Waitlist)

- Building a platform that connects AI prompt creators with users. Creators submit prompts, users generate from them, and revenue is distributed per generation.
- Live in waitlist stage; managing infrastructure independently: DigitalOcean Droplet for backend services, Cloudflare Pages for the frontend.

EARLIER WORK INCLUDES:

- **Order & Event Management System for Studio Senura Seya (University group project)** — Next.js, Spring Boot, MySQL, Tailwind CSS.
- **3D Gaming Room Interior Rendering (University Solo Project)** - 3D gaming room interior using OpenGL and C++ with dynamic lighting systems, texture mapping, interactive camera controls, and real-time animations. [📺](#) [↗](#)
- **AI Challenge Generator** — full-stack coding challenge platform with OpenAI API integration, user auth, and quota management (React · FastAPI). [↗](#) [↗](#)
- **AI Research Agent** — automated developer tool discovery using LangGraph, GPT-4, and Firecrawl. [↗](#)

SKILLS

Immersive / XR: Three.js, React Three Fiber, AR, Colyseus, Unity

Frontend: React, Next.js, TypeScript, Tailwind CSS, HTML5, CSS3

Backend: NestJS, Node.js, Express.js, FastAPI, Spring Boot

Databases: PostgreSQL, MongoDB, MySQL

Cloud & Infrastructure: DigitalOcean, Cloudflare Pages, AWS, Google Cloud

Analytics & SEO: Google Analytics, Google Search Console, Microsoft Clarity, Open Graph / Social Meta

Languages: JavaScript, TypeScript, Python, Java, C++, C#

Soft Skills: Cross-functional collaboration, Agile delivery, Technical problem-solving, Written communication

ADDITIONAL

Languages: English, Sinhala

Certifications: JavaScript Essentials 1 & 2 (Cisco), Google Arcade Facilitator Program (2024), AWS Innovate, AI Agents Fundamentals (Hugging Face)

Activities: Rotaract Club of UOP — Joint Secretary (2023/24), Co-Director of Public Relations (2022/23) · University & School Hockey Team

Writing: Freelance tech writer on Medium (Technology, AI) · RankMath Affiliate Contest Winner (July 2024)